AutoGuardian - Smart Vehicle Anti-Theft System

Pitch Deck for Sponsorship & Support
Cape Town, South Africa

1. Problem Statement

Vehicle theft remains a critical challenge in South Africa, with thousands of cars stolen each year — a reality that affects families, delivery services, and businesses alike. According to the South African Police Service (SAPS), 8611 cars and motorcycles were stolen in the first quarter (April-June) of 2024. This translate to an average of 96 vehicles stolen each day and recovery rates are often low, especially when vehicles are stripped or driven across borders before alerts are raised.

Current anti-theft solutions are failing in key areas:

1. **Lack of Real-Time Feedback**: Many victims only discover their car has been stolen *hours later*, often when it's too late to track or intercept the vehicle.
2. **High Costs**: Advanced vehicle tracking systems often require **monthly subscriptions or high upfront costs —** making them unaffordable for students, low-income individuals, and even small delivery businesses.
3. **Weak Integration**: Most systems don't offer full interaction with **mobile apps** or the ability to remotely control vehicle electronics like engine cut-off or alert toggling.

Real-world scenario:

Imagine Sipho, a delivery driver in Cape Town. After finishing a shift, he parks his company's van outside his home. Overnight, the van is stolen — and the tracking company only reports movement hours later. With no immediate alerts or control, the company loses both the vehicle and the day's inventory. If Sipho had a smart, real-time, mobile-connected anti-theft solution — he could have received a phone alert the moment the car was tampered with, viewed its exact location on a map, and even triggered an emergency kill switch — potentially saving the vehicle.

This is not just a **technology problem**, it's an **economic and emotional one** — with direct impact on **livelihoods, personal safety, and national insurance losses**. The need for a **real-time, affordable, and connected solution** has never been more urgent.

2. The AutoGuardian Solution

AutoGuardian is a smart, affordable, and real-time vehicle anti-theft system designed for the modern South African driver — whether you're a student, parent, fleet manager, or entrepreneur. It combines the power of embedded hardware, mobile apps, and cloud connectivity to provide a comprehensive anti-theft system unlike any on the market.

How It Works:

A small microcontroller-powered device (with GPS, GSM, and sensors) is discreetly installed in your vehicle. The device communicates with a secure cloud server in real time. The owner is instantly notified via the AutoGuardian mobile app if suspicious activity is detected — with the option to track the car live on a map or even shut down the engine remotely via a secure kill switch.

Core Features:

i) **Live Tracking & Location Sharing** - Track your vehicle in real-time on a mobile map interface using GPS + GSM technology.
ii) **Remote Kill Switch** - Instantly disable the vehicle engine from anywhere through the app in case of a theft attempt.
iii) **Motion Detection & Vibration Alerts** - If the vehicle is moved without authorization, the user receives a mobile alert within seconds.
iv) **Tamper Detection** - Alerts are triggered if the system is unplugged, wires are tampered with, or the car is forcibly started.
v) **Mobile App (Flutter-based)** - Simple yet powerful cross-platform app for Android and iOS that connects the user to the system in real time.
vi) **Cloud-Based Event Logging** - All activities (trip logs, alert events, shutdown commands) are stored securely in the cloud for future review or evidence.
vii) **Expandable Architecture** - The system can grow with more features: engine diagnostics (via OBD-II), fuel monitoring, driver behaviour tracking, etc.

Why AutoGuardian is Different:

Problem	**Existing Systems**	**AutoGuardian**
Real-time feedback	Delayed or none	Instant alerts & control
Mobile integration	Limited or clunky	Modern app experience
DIY Installation	Often proprietary	Easy-to-install and customizable
Expandability	Rigid systems	Modular & scalable

3. Key Features

- Real-time location tracking (GPS)
- GSM alerts (theft attempts, disconnection)

- Emergency kill switch via mobile app
- Remote engine shut-off
- JWT-secured REST APIs
- Cloud logging of trip & event data
- Scalable architecture for multi-vehicle fleets

4. System Architecture

The system is made of three key layers:

1. Hardware: ESP32, SIM800L, GPS, relay modules, sensors.
2. Backend: Node.js + MySQL + RESTful API, hosted on Hetzner cloud.
3. Mobile App: Flutter app for Android/iOS to interact with hardware via cloud.

5. Completed Work

- Full backend API with JWT Auth, alerts, GPS logs, and device management
- Sequelize models and migrations
- Live tested cloud backend
- Mobile app with Google Maps integration, device pairing, and alerts display
- Full simulation in Wokwi with all sensors
- Future hardware-ready system for car deployment

6. Hardware Bill of Materials (Per Unit)

- ESP32 Dev Board (microcontroller)
- SIM800L GSM Module
- GPS Module (NEO-6M or similar)
- Relay Module (engine control)
- Buzzer, LED, OLED, push buttons
- Power converters (12V to 5V/3.3V)
- Optional: Gyroscope sensor, touch sensors
- Solderable PCB or protoboard
- Approximate cost: R1700 per car setup

7. Funding Requirements

As a student innovator, I aim to build 1 fully working prototype to test in a real car environment.

Funding Needed: R1700 – for hardware purchase, soldering, and installation materials.

Future Scale: If sponsors support additional kits, the system is ready to scale to 20+

vehicles.

Benefits to Sponsors:
- First-look access to innovation
- Co-branding opportunities
- Early partnership options for resale, licensing, or acquisition

8. Call to Action

I am currently seeking:
- Hardware Sponsorship (R1700 for 1 kit)
- Mentorship or Technical Guidance
- Partnerships with vehicle security firms
- Space to demo or test the system

Help turn this project from simulation to a deployed real-world solution.

9. Contact Information

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